

# **SALES PERFORMANCE & BUSINESS INSIGHTS ANALYSIS USING MYSQL**

**LEVERAGING MYSQL TO OPTIMIZE REVENUE & SUPPLY CHAIN EFFICIENCY**

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## **Project Overview**

This project simulates a real-world sales analytics scenario for a growing retail organization aiming to improve revenue performance and operational efficiency through data-driven decision-making.

The analysis focuses on understanding key revenue drivers, identifying performance gaps across regions, products, and salespersons, and uncovering hidden cost leakages caused by returns and logistics expenses. Instead of limiting the work to SQL queries, the project emphasizes business insights that can support strategic and operational decisions.

The complete analysis was performed using MySQL, starting from data validation and cleaning to multi-dimensional sales, customer, and return analysis. The outcome highlights how structured SQL analysis can help leadership teams improve profitability, optimize sales strategy, and reduce revenue losses.

## **Business Problem Statement**

From a business perspective, the organization faces several challenges:

- Sales performance is uneven across regions, products, and sales teams
- Revenue appears strong at a high level, but profitability is impacted by returns and shipping costs
- Management lacks clear visibility into which areas truly drive growth and which create hidden losses
- Decisions around promotions, inventory, and sales performance are not fully data-backed

The objective of this project is to convert raw transactional data into clear, actionable insights that help management improve revenue quality, not just revenue volume.

## **Dataset Description**

- **Data Source:** Simulated retail transactional sales dataset
- **Type of Data:** Structured sales and order-level data
- **Granularity:** One record per customer order

### **Key Data Attributes:**

- Order details (order date, delivery date, region)
- Product information (product name, quantity, price)
- Salesperson and customer type
- Shipping cost and return status
- Revenue-related calculated fields

The dataset captures the complete sales lifecycle, from order placement to delivery and potential returns.

### **Tools & Technologies Used**

- **Database:** MySQL
- **Query Language:** SQL

### **SQL Concepts Applied:**

- Joins (INNER, LEFT)
- Aggregations and GROUP BY
- Subqueries
- Common Table Expressions (CTEs)
- Window Functions
- Conditional logic (CASE statements)
- Date-based analysis

```
CREATE DATABASE SalesProject;  
USE SalesProject;
```

```
CREATE TABLE sales(  
    OrderDate DATE,  
    Region VARCHAR(50),  
    Product VARCHAR(50),  
    Quantity INT,  
    UnitPrice DECIMAL(10,2),  
    StoreLocation VARCHAR(50),  
    CustomerType VARCHAR(50),  
    Salesperson VARCHAR(50),  
    TotalPrice DECIMAL(10,2),  
    PaymentMethod VARCHAR(50),  
    Promotion VARCHAR(50),  
    Returned INT,  
    OrderID VARCHAR(50) PRIMARY KEY,  
    CustomerName VARCHAR(100),  
    ShippingCost DECIMAL(10,2),  
    DeliveryDate DATE,  
    RegionManager VARCHAR(50)  
);
```

## Data Cleaning & Validation:

```
-- NULL Check
SELECT *
FROM sales
WHERE orderid IS NULL
      OR orderdate IS NULL
      OR product IS NULL;

-- Duplicate Orders
SELECT orderid, COUNT(*)
FROM sales
GROUP BY orderid
HAVING COUNT(*) > 1;
```

```
-- Logical Errors Check
-- Quantity <= 0
SELECT * FROM sales WHERE quantity <= 0;

-- Negative price
SELECT * FROM sales WHERE unit_price < 0;

-- Date Validation
SELECT *
FROM sales
WHERE delivery_date < order_date;
```

- Validated missing and inconsistent values in critical business fields
- Removed duplicate transactions to ensure accurate revenue reporting
- Ensured logical correctness of dates, quantities, and pricing values
- Verified revenue calculations for data consistency
- Excluded returned orders from revenue analysis to prevent overstatement

## Key Analysis & SQL Techniques

- Validated and cleaned transactional data to ensure accuracy before analysis
- Analyzed sales performance across regions, products, and salespersons to identify revenue concentration
- Compared high-performing and low-performing segments to detect growth opportunities
- Evaluated return patterns to understand non-obvious revenue loss drivers
- Assessed shipping costs and returns from a profitability perspective rather than pure sales volume
- Used aggregations and window functions to rank performance and identify trends

The focus throughout the analysis was on **\*\*why performance differs\*\***, not just how much was sold.

## Basic Business KPI

-- Total Revenue

```
SELECT SUM(totalprice) AS total_revenue
FROM sales
WHERE returned = 0;
```

|  | total_revenue     |
|--|-------------------|
|  | 3269902.764500001 |

-- Total Orders

```
SELECT COUNT(DISTINCT orderid) AS total_orders
FROM sales;
```

|  | total_orders |
|--|--------------|
|  | 1500         |

-- Return Rate

```
SELECT
    (SUM(returned) / COUNT(*)) * 100 AS return_percentage
FROM sales;
```

|  | return_percentage |
|--|-------------------|
|  | 24.8000           |

-- Average Order Value (AOV)

```
SELECT
    SUM(totalprice) / COUNT(DISTINCT orderid) AS avg_order_value
FROM sales
WHERE returned = 0;
```

|  | avg_order_value    |
|--|--------------------|
|  | 2898.8499685283696 |

## **Sales Analysis (Region, Product & Salesperson)**

### **Sales Performance Insights**

Sales performance was analyzed across multiple dimensions to identify **what drives revenue and who drives results**.

#### **Regional Performance**

- Revenue is unevenly distributed across regions
- A few regions contribute a disproportionately high share of total sales
- Underperforming regions indicate potential gaps in marketing, distribution, or demand

#### **Product Performance**

- A limited number of products generate the majority of revenue
- High-performing products represent strong market demand
- Low-performing products may require pricing, positioning, or inventory review

#### **Salesperson Performance**

- Significant performance variation exists across salespersons
- Top performers consistently generate higher revenue
- Performance benchmarking can help improve overall team efficiency

This analysis highlights where the company should **double down on strengths and address inefficiencies**.



## Sales Analysis (Product + Region)

-- Region-wise Revenue

```
SELECT region, SUM(totalprice) AS revenue
FROM sales
WHERE returned = 0
GROUP BY region
ORDER BY revenue DESC;
```

|   | region  | revenue           |
|---|---------|-------------------|
| ▶ | North   | 736811.4195000005 |
|   | East    | 683125.228        |
|   | Central | 654013.9330000003 |
|   | South   | 603811.7990000002 |
|   | West    | 592140.3850000001 |

-- Product Performance

```
SELECT product,
       SUM(quantity) AS total_units,
       SUM(totalprice) AS revenue
FROM sales
WHERE returned = 0
GROUP BY product
ORDER BY revenue DESC;
```

|  | product | total_units | revenue            |
|--|---------|-------------|--------------------|
|  | Printer | 1829        | 532599.1095000003  |
|  | Tablet  | 1913        | 510835.135         |
|  | Chair   | 1532        | 471490.76099999998 |
|  | Laptop  | 1704        | 471443.85449999996 |
|  | Monitor | 1537        | 460306.136         |
|  | Desk    | 1621        | 452516.68950000015 |
|  | Phone   | 1489        | 370711.079         |

-- Top 5 Products

```
SELECT product, SUM(totalprice) AS revenue
FROM sales
WHERE returned = 0
GROUP BY product
ORDER BY revenue DESC
LIMIT 5;
```

|  | product | revenue            |
|--|---------|--------------------|
|  | Printer | 532599.1095000003  |
|  | Tablet  | 510835.135         |
|  | Chair   | 471490.76099999998 |
|  | Laptop  | 471443.85449999996 |
|  | Monitor | 460306.136         |

## Customer & Salesperson Analysis

-- Retail vs Wholesale

```
SELECT customertype,  
       COUNT(orderid) AS orders,  
       SUM(totalprice) AS revenue  
FROM sales  
WHERE returned = 0  
GROUP BY customertype;
```

| customertype | orders | revenue            |
|--------------|--------|--------------------|
| Wholesale    | 573    | 1628793.1269999994 |
| Retail       | 555    | 1641109.6375000002 |

-- Top Salesperson

```
SELECT salesperson,  
       SUM(totalprice) AS revenue  
FROM sales  
WHERE returned = 0  
GROUP BY salesperson  
ORDER BY revenue DESC;
```

|   | salesperson | revenue           |
|---|-------------|-------------------|
| ▶ | Bob         | 606773.9835000001 |
|   | Alice       | 591860.9474999994 |
|   | Frank       | 555391.8790000003 |
|   | Eva         | 531819.4495000001 |
|   | Carlos      | 496695.5805000002 |
|   | Diana       | 487360.9244999999 |

-- Manager Performance

```
SELECT regionmanager,  
       SUM(totalprice) AS revenue  
FROM sales  
WHERE returned = 0  
GROUP BY regionmanager;
```

|   | regionmanager | revenue           |
|---|---------------|-------------------|
| ▶ | Eric          | 683125.228        |
|   | Sophie        | 603811.7990000002 |
|   | Ryan          | 736811.4195000005 |
|   | Cameron       | 654013.9330000003 |
|   | Wendy         | 592140.3850000001 |

-- Monthly Trend

SELECT

YEAR(orderdate) AS year,  
MONTH(orderdate) AS month,  
SUM(totalprice) AS revenue

FROM sales

WHERE returned = 0

GROUP BY year, month

ORDER BY year, month;

| year | month | revenue            |
|------|-------|--------------------|
| 2023 | 1     | 132179.5695        |
| 2023 | 2     | 117238.8515        |
| 2023 | 3     | 142531.1215        |
| 2023 | 4     | 98612.3645         |
| 2023 | 5     | 71813.475          |
| 2023 | 6     | 102020.74049999999 |
| 2023 | 7     | 93366.61349999996  |
| 2023 | 8     | 132966.75700000004 |
| 2023 | 9     | 110702.1           |
| 2023 | 10    | 44922.2075         |
| 2023 | 11    | 113050.00850000001 |
| 2023 | 12    | 98552.888          |
| 2024 | 1     | 160871.9045        |
| 2024 | 2     | 96288.86799999999  |
| 2024 | 3     | 102082.56100000002 |
| 2024 | 4     | 139187.64200000002 |
| 2024 | 5     | 70128.3885         |
| 2024 | 6     | 114499.091         |
| 2024 | 7     | 103818.602         |
| 2024 | 8     | 101914.41350000001 |
| 2024 | 9     | 105390.46899999998 |
| 2024 | 10    | 127054.54049999997 |

result 12 x

-- Yearly Sales Trend

SELECT YEAR(orderdate) AS year,  
SUM(totalprice) AS revenue

FROM sales

WHERE returned = 0

GROUP BY year

ORDER BY year;

|   | year | revenue            |
|---|------|--------------------|
| ▶ | 2023 | 1257956.6970000006 |
|   | 2024 | 1345291.084        |
|   | 2025 | 666654.9835000004  |

## Profit & Cost Thinking

-- High Cost Orders

```
SELECT orderid, shippingcost
FROM sales
ORDER BY shippingcost DESC
LIMIT 10;
```

|   | orderid   | shippingcost |
|---|-----------|--------------|
| ▶ | REG100953 | 49.98        |
|   | REG100386 | 49.95        |
|   | REG100142 | 49.93        |
|   | REG100774 | 49.93        |
|   | REG101266 | 49.93        |
|   | REG101120 | 49.91        |
|   | REG100646 | 49.91        |
|   | REG100389 | 49.9         |
|   | REG101102 | 49.82        |
|   | REG100363 | 49.79        |

-- Shipping Cost Impact

```
SELECT
    SUM(shippingcost) AS total_shipping_cost,
    SUM(totalprice) AS total_revenue
FROM sales;
```

|   | total_shipping_cost | total_revenue      |
|---|---------------------|--------------------|
| ▶ | 41260.94000000011   | 4379992.4284999985 |

# Return Analysis (The “Why” Behind Revenue Loss)

## Return & Loss Analysis

Returns represent a hidden cost that directly impacts profitability. This analysis focused on identifying **patterns behind returned orders**:

- Specific products show higher return frequency
- Certain promotions are linked with increased returns
- Regional differences suggest operational or logistics issues

These findings indicate that returns are not random but **systematic**, offering opportunities for:

- Product quality improvement
- Better customer expectation management
- Optimized promotion strategies

Reducing return rates can significantly improve net revenue without increasing sales volume.

# Final Conclusion & Strategic Recommendations

## Conclusion & Recommendations

This project demonstrates how structured data analysis can uncover **deep business insights from transactional data**.

### Key Conclusions

- Revenue concentration creates both strength and risk
- High-performing products and regions drive most of the business value
- Returns and shipping costs are key profit leakage points
- Seasonal trends strongly influence sales performance

### Strategic Recommendations

- Invest more aggressively in high-performing regions and products
- Use data-driven forecasting for inventory and supply chain planning
- Review promotions linked with high return rates

Implement performance-based coaching for sales teams